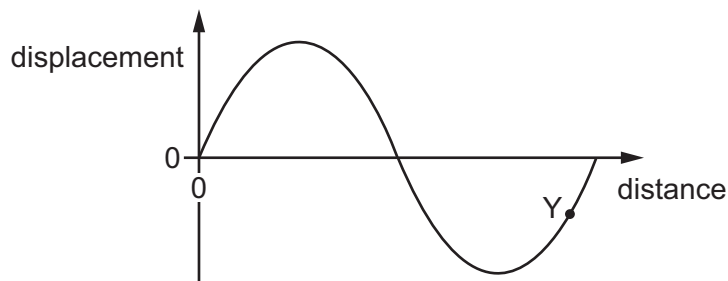


- 25** A source emits a progressive sound wave in the horizontal direction. The wave travels away from the source in a direction towards the right.

The graph shows the variation of the displacement of the particles from their equilibrium positions with distance from the source at a particular instant in time. Displacements to the right of the equilibrium positions are shown as positive.



Position Y represents the displacement of a particle in the sound wave at a particular distance from the source at this instant.

Which statement about the motion of this particle is correct?

- A** The particle is moving away from the source.
 - B** The particle is moving towards the source.
 - C** The particle is moving upwards.
 - D** The particle is moving downwards.
- 26** Two polarising filters are placed next to each other so that their planes are parallel.

The first polarising filter has its transmission axis at an angle of 50° to the vertical.

The second polarising filter has its transmission axis at an angle of 20° to the vertical. The angle between the transmission axes of the two polarising filters is 30° .

A beam of vertically polarised light of intensity 8.0 W m^{-2} is incident normally on the first polarising filter.

What is the intensity of the light that is transmitted from the second polarising filter?

- A** zero
 - B** 2.5 W m^{-2}
 - C** 2.9 W m^{-2}
 - D** 6.0 W m^{-2}
- 27** A dolphin is swimming at a speed of 16.0 m s^{-1} directly towards a stationary underwater microphone.

The dolphin emits a sound of frequency 122 kHz . The speed of sound in water is 1480 m s^{-1} .

What is the frequency of sound detected by the microphone?

- A** 1300 Hz
- B** 1330 Hz
- C** 121 kHz
- D** 123 kHz